

Structural Entropy: A Decade Survey of Theory, Algorithms, Learning Paradigms, and Applications

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Abstract

Classical information theory is fundamentally limited by its local perspective, analyzing pairwise interactions while ignoring the larger hierarchical architecture of complex systems. *Structural Entropy* (SE) presents a paradigm shift: it extends Shannon entropy to quantify information on a global scale and measures the uncertainty embedded in a system’s organizational hierarchy. Since its introduction in 2016, SE has evolved from a principled measure for community detection into a broadly applicable framework spanning graph learning, reinforcement learning, bioinformatics, social-network forensics, and pattern recognition—yet a systematic, unified synthesis of its theory, algorithms, and applications has been lacking.

This survey provides a comprehensive decade-long overview of SE research. We (i) formalize its theoretical foundations, covering the seminal framework, key extensions to directed, dynamic, and multi-relational graphs, and formal connections to spectral entropy; (ii) survey the computational landscape, cataloguing heuristic, constrained, and differentiable SE minimization algorithms together with their complexity profiles; (iii) analyze SE-driven learning paradigms in graph pooling, structure augmentation, and reinforcement learning; (iv) review cross-domain applications in bioinformatics, transportation, geoscience, social networks, and pattern recognition; and (v) present a reproducibility benchmark comparing SE-based objectives against competing graph-clustering methods. We further discuss open research challenges and future directions, including unified frameworks for heterogeneous and hypergraph structures, scalable optimization, and the integration of SE into large language model pipelines. A companion open-source library, `glass-jax`, accompanies this survey as a reproducibility artifact.

Keywords: structural entropy, graph clustering, hierarchical community detection, graph neural networks, reinforcement learning, information theory.

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1 Introduction

Information lies at the heart of artificial intelligence. From Shannon’s entropy [48] guiding learning algorithms to the Information Bottleneck [56] shaping representations, virtually every paradigm in AI can be viewed as compressing, transferring, organizing, or extracting meaning from information. Yet, while much of classical information theory is concerned with quantities of uncertainty at a *local* or *pairwise* level, **Structural Entropy (SE)** [16] offers a fundamentally distinct and complementary perspective: it captures the *global structure* and *hierarchical uncertainty* embedded within complex information systems.

At its core, SE provides a rigorous method to quantify and decode the uncertainty embedded in the architecture of a complex system, modelled as a graph or matrix. Its significance for the AI community is twofold:

- **Metric.** SE serves as a fundamental metric that quantifies a system’s intrinsic uncertainty. Low SE signals a clear, efficient hierarchical organization—implying functional coherence and robustness. High SE indicates a system that is difficult to decompose, approaching the behaviour of a random graph. This analytical power is exploited in privacy-preserving graph analysis (deliberately increasing SE to obscure community structure [32]), knowledge quantification [63], and AI safety evaluation [86].
- **Optimization principle.** SE provides a principled method to *discover* a system’s optimal hierarchical structure. Minimizing SE decodes the most efficient information-theoretic abstraction of a system, revealing a nested, hierarchical partitioning of its components—the “true” community structure [16] or most effective decision hierarchy [91].

In the context of *Transactions on Graph Intelligence and Network Applications* (TGINA), SE is timely because modern graph intelligence systems increasingly need more than accurate predictions: they need interpretable, auditable, and reusable structural abstractions. This is especially visible in graph-enhanced large language model pipelines, including GraphRAG-style retrieval, knowledge-graph reasoning, tool-use memory, and autonomous-agent interaction graphs. These systems generate relational structures whose trustworthiness depends on whether meaningful hierarchies can be separated from incidental links. SE provides a principled way to measure such organization and, through minimization, to construct candidate hierarchies for retrieval, reasoning, and decision making.

Motivation: Why a Decade Survey Now?

The foundational SE paper [16] was published a decade ago. In the intervening years, SE-based methods have proliferated across graph pooling [66, 45], data augmentation [65, 64], graph structure learning [102, 7], reinforcement learning [91, 88, 85], bioinformatics [97, 4, 21], social-network analysis [41, 72, 3], and pattern recognition [101, 81, 61]. Despite this rapid proliferation, a comprehensive and systematic review consolidating these advances has been lacking.

This article aims to fill that critical gap. The present work substantially extends our earlier conference survey [51] published at IJCAI 2025:

1. **Expanded theory.** We provide a self-contained, rigorous treatment of SE’s mathematical foundations, including proofs of key properties and formal comparisons with alternative graph-complexity measures (modularity, map equation, von Neumann entropy).
2. **Comprehensive algorithm catalogue.** We survey all known SE minimization algorithms, extending the conference table with complexity analysis, constraint categories, and reproducibility notes.
3. **Reproducibility benchmark.** We present a new empirical evaluation comparing SE-based graph clustering objectives against Louvain, Infomap, and differentiable baselines across multiple datasets and random seeds using the `glass-jax` companion library.
4. **Expanded application coverage.** We include additional application domains (speech processing, geoscience knowledge quantification, LLM integration) and updated literature through 2026.
5. **Open problems.** We provide a structured discussion of open research challenges and future directions that were absent from the conference version.

Paper Organization

The remainder of this survey is organized as follows. Section 2 positions SE against modularity, flow-based, spectral, and deep graph-clustering methods. Section 3 introduces a taxonomy of SE research and its formal definitions. Section 4 covers theoretical foundations. Section 4 surveys computational algorithms for SE minimization. Section 5 examines SE-driven learning paradigms (graph learning and reinforcement learning). Section 5.3 reviews cross-domain applications. Section 6 presents the reproducibility benchmark. Section 8 discusses open problems and future directions. Section 9 concludes.

2 Related Work

Structural Entropy is part of a broader family of graph-complexity and graph clustering methods, but it differs from most classical objectives in treating hierarchical organization as the primary object to be decoded. We summarize the closest methodological neighbors below and use them as reference points for the taxonomy and benchmark sections.

Modularity-based community detection. Modularity compares observed within-community edge density against a configuration-model null graph [33]. Louvain and Leiden-style optimization methods make this objective highly scalable, with Leiden additionally improving partition connectivity and refinement stability [57]. These methods are strong default baselines for topology-driven clustering, but their flat partitions and known resolution effects make them less suited to explicitly nested abstractions. SE instead evaluates the coding cost induced by an encoding tree, so the partition is interpreted as an information hierarchy rather than only a density surplus.

Flow-based compression. The map equation and Infomap encode random-walk trajectories using module codebooks [47]. They are conceptually close to SE because both

explain graph organization through compression, but the objects being compressed are different: Infomap compresses flows, while SE compresses structural uncertainty under volumes, cuts, and tree refinement. This distinction matters in applications where the observed graph represents an information system rather than an explicit navigation process.

Spectral partitioning. Spectral clustering and normalized cuts use eigenvectors of graph Laplacians to expose low-conductance partitions [49, 34]. These methods provide strong relaxations for balanced graph cuts and remain important for initialization and diagnostics. However, spectral objectives are usually flat and externally parameterized by the desired number of clusters. SE can use spectral information, as in the VNE connection, but its target is a minimum-description hierarchy rather than an eigenspace partition alone.

Deep graph clustering and pooling. Neural graph clustering methods such as Diff-Pool, MinCutPool, and DMoN learn soft assignments inside GNN architectures [78, 1, 58]. They are effective when node attributes and task labels provide additional semantic signal, but their objectives are often architectural regularizers rather than standalone measures of structural organization. Recent SE-based neural methods, including LSENet and DeSE, bridge this gap by making structural entropy differentiable [53, 94]. The journal contribution of this survey is to place these neural methods in the same conceptual map as the original combinatorial SE algorithms and their domain-specific extensions.

3 Formal Foundations and Taxonomy

3.1 Basic Mathematical Constructs

3.1.1 Encoding Tree of a Finite Set

Let A be a finite set representing the entities in a complex system (e.g., nodes in a network or data points in a dataset). A natural model for the structure of such a system is a *hierarchical abstraction* in which elements are recursively grouped into nested subsets.

Definition 3.1 (Encoding Tree). *Given a finite set A , an encoding tree of A is a rooted tree T such that: (1) the root λ corresponds to the entire set $T_\lambda = A$; (2) for every non-leaf node $\alpha \in T$ with children $\beta_0, \dots, \beta_l$ the corresponding subsets form a partition $T_\alpha = \bigsqcup_{j=0}^l T_{\beta_j}$; (3) each leaf $\gamma \in T$ corresponds to a singleton $T_\gamma = \{a\}$.*

The *dimension* of an encoding tree is its depth. The extreme case—an encoding tree of depth 1 where the root is directly partitioned into singleton leaves—yields the **1-dimensional SE**, which reduces to Shannon entropy.

3.1.2 Structural Entropy

Consider a non-negative, irreducible matrix $A \in \mathbb{R}_{\geq 0}^{n \times n}$ representing an information system over n elements indexed by $V = \{1, \dots, n\}$. Let $\boldsymbol{\pi} = (\pi_1, \dots, \pi_n)$ be the stationary distribution of the Markov chain derived from A by row normalization, with transition probability $p_{xy} = \pi_x b_{xy}$ where $b_{xy} = a_{xy} / \sum_j a_{xj}$.

For a subset $X \subseteq V$ define:

$$V_X = \sum_{x \in X} \pi_x, \quad p_X = \sum_{y \notin X} \sum_{x \in X} p_{yx}, \quad q_X = \sum_{x \in X} \sum_{y \notin X} p_{xy}.$$

Definition 3.2 (Structural Entropy under a Tree). *The structural entropy of A under encoding tree T is*

$$\mathcal{H}^T(A) = - \sum_{\substack{\alpha \in T \\ \alpha^- \neq \lambda}} p_\alpha \log_2 \frac{V_\alpha}{V_{\alpha^-}},$$

where α^- denotes the parent of α in T .

Definition 3.3 (Structural Entropy of a System). *The structural entropy of an information system is*

$$\mathcal{H}(A) = \min_T \mathcal{H}^T(A).$$

The minimizing tree T^* encodes the system’s most efficient information-theoretic abstraction—its natural hierarchical community structure [16].

3.2 Relationship to Other Graph-Complexity Measures

Three major graph-complexity measures are related to SE:

Modularity [33]. Modularity Q measures the fraction of edges within communities minus the expected fraction under a null model. While modularity seeks to maximize within-community density relative to a random graph, SE minimization seeks to *minimize informational uncertainty* within a hierarchical partition. Modularity is resolution-limited (it cannot detect communities smaller than a scale-dependent threshold); SE is not subject to the same limitation.

Map Equation [47]. Infomap minimizes the expected description length of a random walk under a two-level codebook. SE generalization replaces the two-level restriction with an arbitrary tree, and replaces flow coding with a volume-and-cut measure. Formally, [31] proved that for any undirected graph $\mathcal{H}(A) \leq L^*(A) \leq \mathcal{H}(A) + \gamma$, where L^* is the optimal map-equation description length and γ depends only on the graph’s degree sequence.

Von Neumann Entropy (VNE). VNE is the entropy of the normalized graph Laplacian’s eigenvalue distribution. [31] established that $\mathcal{H}(A) - \text{VNE}(A) \in [0, \gamma]$, providing the first formal spectral interpretation of SE.

3.3 Taxonomy of SE Research

Figure 1 provides a high-level taxonomy organizing SE research into four pillars—*Theory*, *Computational Algorithms*, *Learning Methods*, and *Cross-Domain Applications*—each containing major sub-themes and representative works. We follow this taxonomy throughout the survey.

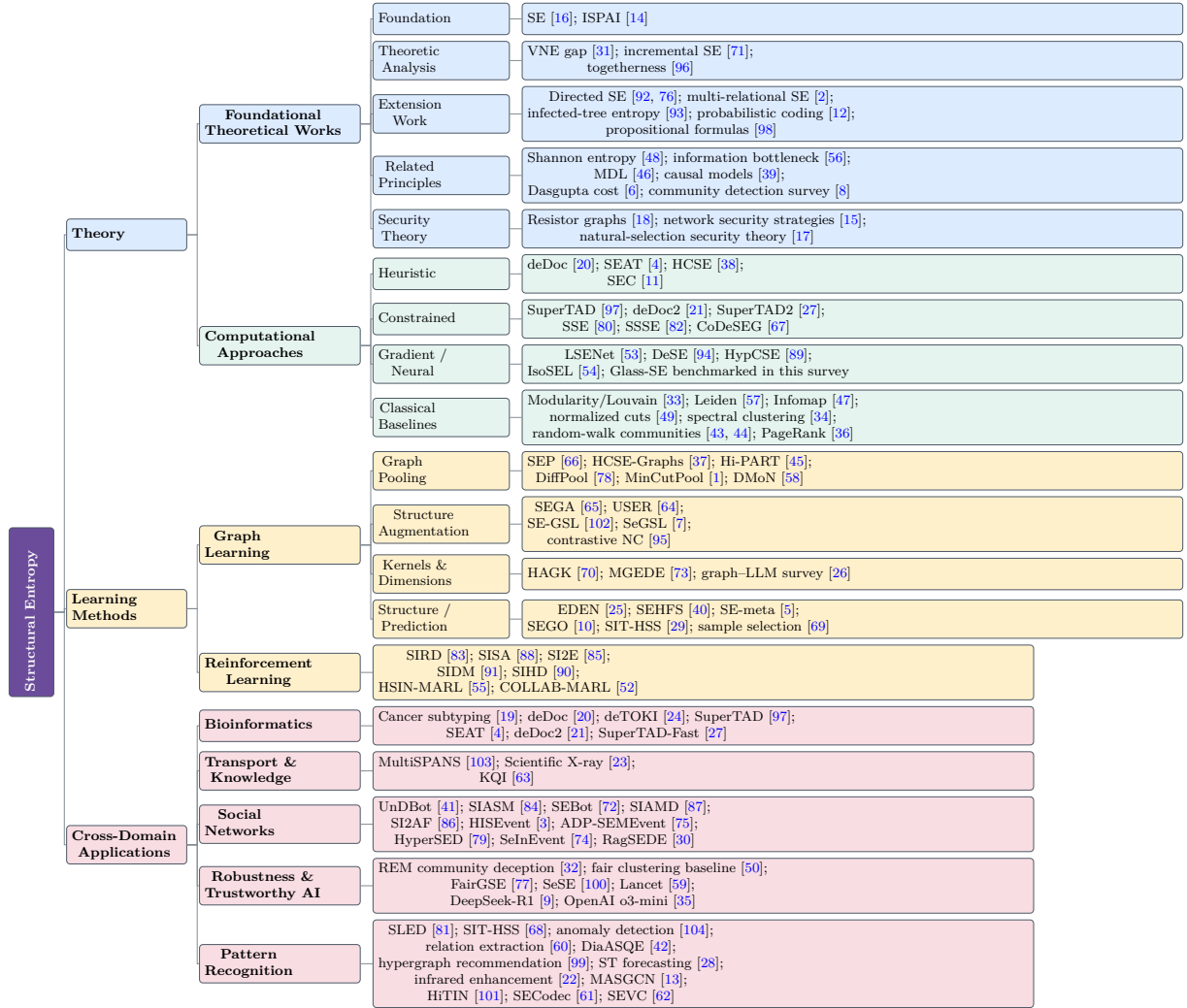


Figure 1: Taxonomy of Structural Entropy research for the TGINA journal version (2016–2026), following the vertical visual organization of the IJCAI conference taxonomy [51] while adding journal-version categories and citations.

4 Theoretical Foundations and Computational Algorithms

4.1 Foundational Theoretical Works

Foundation Theory. The concept of SE was introduced by [16] as a principled, information-theoretic framework for quantifying the complexity of a graph’s structure. The seminal work defined SE as the minimum code length required to describe a network’s hierarchy under a random walk process, bridging graph theory with Shannon’s information principles. It established that minimizing SE reveals a graph’s intrinsic multi-scale organization. Building on this, [14] significantly broadened the theoretical landscape by situating SE within the *mathematical principles of the information world*, formulating fundamental information laws and the Information Science Principles of AI (ISPAI), providing axiomatic foundations for a comprehensive structural information theory.

Theoretical Analysis. Subsequent research has elucidated SE’s properties and forged connections with established graph-theoretic concepts. The “togetherness” metric [96] uses SE reduction to quantify the efficacy of network integration when adding inter-community edges. [31] established a significant theoretical link between SE and spectral graph theory: for any undirected graph, the difference between SE and the von Neumann graph entropy (VNE) is bounded within the interval $[0, \gamma]$, formally connecting SE to spectral properties of the graph Laplacian. [71] proposed incremental computation paradigms for dynamic graphs, enabling efficient encoding tree updates as the network evolves.

Extensions. The original formulation targets simple graphs; significant research has extended SE to more complex network structures. For directed graphs, [76] introduced a localized 2D SE via a flow-based algorithm, while [92] proposed a random walk-based entropy (RWE) with a proven theoretical connection to VNE. Multi-relational SE [2] uses a random surfing model that jointly optimizes node and relation selection. The *infected tree entropy* (ITE) [93] specializes SE for source localization in information cascades. SE-guided probabilistic coding [12] regularizes embeddings in natural language understanding tasks.

4.2 Computational Algorithms

Table 1 summarizes all known SE minimization algorithms. We organize them by optimization strategy.

Heuristic Methods. deDoc [20] pioneered SE minimization for genomic structure detection with a greedy merging strategy. SEAT [4] adapts this for single-cell omics by applying SE minimization on cell–cell graphs to reveal nested subpopulations. HCSE [38] is a widely used, general-purpose hierarchical clustering framework that recursively isolates the sparsest levels of a hierarchy through SE-guided optimization, automatically determining hierarchy depth without hyperparameters.

Constrained Optimization. SuperTAD [97] combines interval dynamic programming with SE minimization under *spatial contiguity* constraints. SuperTAD2 [27] introduces

Table 1: Summary of SE minimization algorithms. n : nodes; m : edges; t : iterations; k : flat clusters; h : arbitrary tree depth.

Algorithm	Levels	Constraint	Time Complexity	Code	Lang
HCSE [38]	h	Unconstrained	$O(n^2)$	Link	Python
deDoc [20]	2/3	Unconstrained	$O(n \log^2 n)$ or $O(n^2)$	Link	Java
deDoc2 [21]	h	Unconstrained	$O(n^2)$	Link	Java
SSE [80]	h	Label, Pairwise	$O(h(m \log n + n))$	Link	Python
SSSE [82]	h	Label, Pairwise	$O(n \log^2 n + nkt)$	Link	Python
SuperTAD [97]	h	Contiguity	$O(n^4 k^2 h)$	Link	C++
SuperTAD2 [27]	h	Contiguity	$O(n^4 k^2 h)$	Link	C++
PYSEAT [4]	h	Contiguity	$O(n \log n)$	Link	Python
LSENet [53]	2	Unconstrained	$O(nm)$	Link	Python
DeSE [94]	2	Unconstrained	$O(nm)$	—	Python
CoDeSEG [67]	2	Overlapping	$O(nt)$	Link	C++

matrix discretization and approximation strategies achieving $\approx 10\times$ speedup. deDoc2 [21] extends the contiguity approach to single-cell analysis to resolve chromatin hierarchies across the cell cycle. SSE [80] unifies *semi-supervised* constraints (must-link / cannot-link pairwise constraints and label constraints) by reformulating SE to penalize violations during encoding tree construction. SSSE [82] scales this to massive datasets via graph sampling and incremental cluster insertion, reducing time complexity from quadratic to nearly linear.

Differentiable / Gradient-Based Methods. LSENet [53] creates a differentiable SE formulation in hyperbolic space, enabling cluster-free hierarchical partitioning via backpropagation. DeSE [94] pioneers a *soft assignment structural entropy* allowing co-design of graph topology and cluster assignments in an end-to-end model. CoDeSEG [67] offers a game-theoretic formulation for near-linear-time overlapping community detection.

Complexity and Trade-off Analysis. Heuristic approaches offer scalability and generality but are limited in their ability to incorporate domain knowledge. Constrained methods integrate domain-specific knowledge at the cost of broader applicability. Differentiable methods bridge SE and deep learning but introduce hyperparameter sensitivity and potential interpretability concerns. A key open challenge is hybridizing these paradigms to achieve an optimal balance of scalability, accuracy, and interpretability, especially for dynamic or heterogeneous networks (see Section 8).

5 Structural Entropy in Graph Learning and Reinforcement Learning

5.1 Graph Learning Paradigms

SE has emerged as a powerful information-theoretic tool for graph representation learning, offering a principled, quantitative measure of a graph’s inherent hierarchical organization.

5.1.1 Graph Pooling

In graph-level tasks, pooling operations are essential for downsampling nodes to create compact representations of entire graphs. Traditional methods often suffer from information loss by applying stepwise local pooling, which destroys the larger community structure. *SE Pooling* (SEP) [66] addresses this by using global SE minimization to generate all cluster assignments simultaneously, preserving the multi-level community map. *Hi-PART* [45] constructs Hierarchical Partition Trees using SE, explicitly modelling the nested, multi-scale nature of graph structures and surpassing the expressive power of the standard Weisfeiler-Lehman test. *EDEN* [25] further extends SE-based pooling to dynamic and evolving graphs.

5.1.2 Structure Augmentation

Graph structure augmentation underpins modern self-supervised learning on graphs. Random augmentations risk discarding crucial information; SE provides a principled way to guide this process by identifying and preserving the graph’s intrinsic low-entropy structure. *SEGA* [65] constructs a stable anchor view by finding an encoding tree with minimal SE, filtering noise while maintaining the essential community hierarchy for contrastive learning. *USER* [64] incorporates SE directly into its objective function to learn an “innocuous graph” robust to perturbations. *SE-GSL* [102] uses a three-stage optimization: maximizing 1D SE to enhance information content, minimizing high-dimensional SE to abstract the graph into hierarchical trees, then using deduction SE to guide adaptive graph reconstruction. *SeGSL* [7] extends this by using SE-derived minimal-tree structures to guide multi-view fusion based on community influence.

5.1.3 Graph Kernels and Embedding Dimensions

HAGK [70] leverages SE to generate multi-level, abstract representations of each graph’s intrinsic organization for use in a graph kernel that measures similarity between entire graphs. *MGEDE* [73] calculates a high-level SE from multi-order adjacency matrices and minimizes it alongside attribute entropy, explicitly determining the most suitable number of embedding dimensions.

5.2 Reinforcement Learning Integration

Integrating SE into reinforcement learning (RL) equips agents with a mechanism to perceive and decompose complex problems, moving beyond flat state-action spaces toward a structured understanding of their environment.

5.2.1 State and Action Abstraction

The *SISA* [88] framework uncovers hierarchical state structure by optimizing an encoding tree from a state graph using SE, creating an efficient and task-relevant state representation. *SIRD* [83] reframes role discovery in multi-agent RL as hierarchical action space clustering via SE minimization on an action graph.

5.2.2 Intrinsic Motivation and Exploration

SI2E [85] formulates an intrinsic reward by maximizing value-conditional SE: agents are encouraged to explore states where their actions lead to the greatest reduction in structural uncertainty. *SIDM* [91] utilizes SE at multiple stages to structure the entire decision-making and skill-discovery pipeline. *COLLAB-MARL* [52] uses SE to quantify interaction complexity in multi-agent systems, revealing emergent coalition structures by minimizing the SE of a dynamic agent interaction graph.

5.2.3 Discussion

SE integration into RL provides mathematical tools to build a *cognitive architecture* for agents: parsing their world into a hierarchy of concepts, skills, and goals. *SISA* and *SIRD* demonstrate how SE can automate the discovery of abstractions that have historically required significant human engineering. *SI2E* shows that exploration need not be random—it can be a deliberate process of *information seeking* aimed at resolving structural uncertainty. Key future directions include scaling these methods to partially observable environments, meta-RL, and curriculum learning.

5.3 Cross-Domain Applications

5.3.1 Bioinformatics

SE-based genomic analysis began with cancer subtyping [19], correlating entropy-minimized gene expression patterns with clinical outcomes. The first SE-based TAD detector, *deDoc* [20], identified large-scale topologically associating domains from Hi-C data. *SuperTAD* [97] employed dynamic programming to resolve hierarchical chromatin domains; *SuperTAD2* [27] significantly accelerated this via matrix discretization. At single-cell resolution, *SEAT* [4] minimizes global uncertainty in cell-cell graphs to improve pseudo-time inference. *deTOKI* [24] analyzes regulatory dynamics at the single-cell level, and *deDoc2* [21] resolves cell-cycle-dependent chromatin hierarchies.

5.3.2 Transportation and Geoscience

MultiSPANS [103] improves traffic forecasting by using SE minimization to optimize spatial attention mechanisms, generating hierarchical encoding trees of road networks. The “Scientific X-ray” method [23] constructs “idea trees” from citation networks to identify high-potential research directions. *KQI* [63] redefines knowledge quantification by measuring hierarchical disorder differences in citation networks, producing a Knowledge Quantification Index that identifies overlooked but influential works and Nobel Prize-winning topics.

5.3.3 Social Networks

UnDBot [41] offers unsupervised bot detection via heterogeneous SE on multi-relational graphs. *SEBot* [72] integrates SE with multi-view contrastive learning for adversarially robust detection. *SIASM* [84] models how social bots optimally select followers using conditional SE minimization. *SI2AF* [86] evaluates news-detection system robustness using an SE-based influence metric that combines content relevance with community structure. For social event detection, *HISEvent* [3] incrementally builds message graphs with 1D SE and detects events using 2D SE optimization; *ADP-SEMEvent* [75] integrates adaptive differential privacy; *HyperSED* [79] uses hyperbolic-space differentiable SE-guided partitioning.

5.3.4 Pattern Recognition

In *visual computing*, *SLED* [81] detects skin lesions by minimizing the entropy of multi-scale superpixel graphs. *SIT-HSS* [68] performs hierarchical superpixel segmentation via iterative SE-guided merging.

In *natural language processing*, *HiTIN* [101] injects the syntactic hierarchy of sentences via SE into graph-based representations for hierarchical text classification. *MASGCN* [13] applies an SE-based loss to guide a GCN for aspect-based sentiment analysis.

In *speech processing*, *SECodec* [61] applies SE to cluster feature nodes for efficient compressed speech representation. *SEVC* [62] uses 2D SE to group reference speech frames into semantic clusters for high-fidelity voice conversion.

6 Reproducibility Benchmark

6.1 Experimental Protocol

To complement our survey with empirical evidence, we present a reproducibility benchmark using the `glass-jax` companion library—a JAX-based differentiable graph clustering framework that implements soft versions of Modularity, the Map Equation, and 2D Structural Entropy (see Appendix A for library details). All results are reported as **mean $\pm$ 95% confidence interval** over 5 random seeds.

Datasets.

- **Karate**: Zachary’s Karate Club ($N=34$, $K=2$).
- **Caveman(10 $\times$ 10)**: Connected Caveman graph of 10 cliques of size 10 ($N=100$, $K=10$).
- **SBM (Clean)**: Stochastic Block Model ($N=150$, $K=3$, $P_{\text{in}}=0.30$, $P_{\text{out}}=0.05$).
- **SBM (Noisy)**: Stochastic Block Model ($N=150$, $K=3$, $P_{\text{in}}=0.15$, $P_{\text{out}}=0.08$; signal near the community-detection threshold).

Algorithms compared. We include two classes of baselines:

Non-SE baselines (community detection):

1. **Louvain** [33]: official `python-louvain` implementation.
2. **Leiden**: improved Louvain with guaranteed connectivity.
3. **Infomap** [47]: official C++ Python wrapper.
4. **Spectral**: normalized spectral clustering with k -means on the top- K Laplacian eigenvectors.

SE-based methods:

5. **deDoc** [20]: the founding SE algorithm; greedy agglomerative merging that minimises 2D structural entropy at each step (`agglomerative_se_clustering` in the companion library). The reference implementation is $O(N^3)$; results are reported only for Karate ($N=34$); larger datasets are marked $-$.
6. **HCSE** [38]: hierarchical SE clustering. Results here use the companion library’s `hierarchical_se_clustering`, which builds a coding tree from multistart base labels and extracts a flat cut.
7. **SEClust-Auto***: `multistart_se_heuristic` (8 restarts, free K) from the companion library.
8. **SEClust-CK***: `constrained_k_multistart`; multistart SE local search with K fixed to the ground-truth value.
9. **SEClust-ML***: `multilevel_se_clustering`; multi-level coarsen-and-refine SE optimizer (free K).
10. **SEP** [66]: greedy 2D SE pooling algorithm.
11. **LSENet** [53]: differentiable SE via neural network projection.
12. **Glass-Mod**: soft Modularity maximization (`glass-jax`).
13. **Glass-Map**: soft Map Equation minimization (`glass-jax`).
14. **Glass-SE**: soft 2D SE minimization (`glass-jax`).

* From the `glass-jax` companion library (this survey).

All `glass-jax` methods use a multi-start `vmap` engine: four parallel trajectories (1 spectral + 3 random initializations) over 500 Adam steps ($\eta = 0.05$) with temperature annealing from $\tau = 1.0$ to $\tau = 0.01$.

Algorithms excluded from this benchmark. Three SE methods from Table 1 require a fundamentally different evaluation protocol and are therefore excluded. (i) **SSE** [80] is *semi-supervised*: it takes a subset of ground-truth node labels as input, so evaluating it in an unsupervised protocol would be misleading. (ii) **CoDeSEG** [67] detects *overlapping* communities; the standard ARI/NMI metrics for disjoint partitions are not applicable, and the method requires overlapping-specific metrics (F1-overlap, NMI-overlap). (iii) **DeSE** [94] jointly trains topology and assignments end-to-end; no standalone graph-partition interface or public code is available at the survey submission date. **Super-TAD** [97] additionally requires a Hi-C contiguity constraint not present in generic graph benchmarks. **deDoc** is included above for Karate only; the reference implementation is $O(N^3)$ and infeasible for $N > 50$.

6.2 Results: Clustering Accuracy

Table 2 shows ARI and NMI for all algorithms on all datasets.

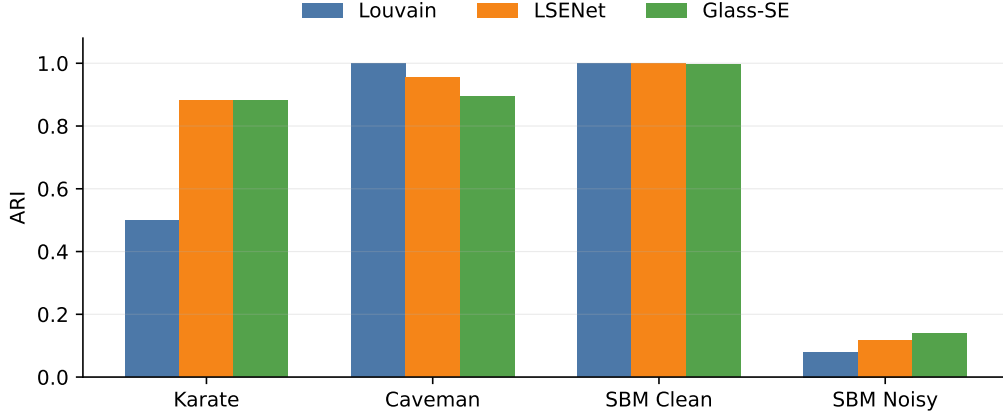


Figure 2: Representative ARI comparison across synthetic and small real-world topological benchmarks. Glass-SE is competitive with strong baselines on clean and noisy topology, while feature-augmented LSENet is strongest when semantic features align with labels.

Table 2: Clustering accuracy (ARI and NMI, mean $\pm$ 95% CI over 5 seeds). Higher is better. Bold: best per dataset. †: SE-based method. *: companion library (this survey). –: method infeasible at this scale ($O(N^3)$, $N>50$).

Dataset	Algorithm	ARI	NMI
Karate ($N=34$)	Louvain	0.500 $\pm$ 0.010	0.597 $\pm$ 0.003
	Leiden	0.500 $\pm$ 0.010	0.597 $\pm$ 0.003
	Infomap	0.684 $\pm$ 0.000	0.691 $\pm$ 0.000
	Spectral	0.620 $\pm$ 0.040	0.650 $\pm$ 0.030
	deDoc†	0.434 $\pm$ 0.000	0.571 $\pm$ 0.000
	HCSE†	0.882$\pm$0.000	0.837$\pm$0.000
	SEClust-Auto†*	0.337 $\pm$ 0.052	0.516 $\pm$ 0.029
	SEClust-CK†*	0.882$\pm$0.000	0.837$\pm$0.000
	SEClust-ML†*	0.393 $\pm$ 0.069	0.546 $\pm$ 0.041
	SEP†	0.882$\pm$0.000	0.837$\pm$0.000
	LSENet†	0.882$\pm$0.000	0.837$\pm$0.000
	Glass-Mod	0.882$\pm$0.000	0.837$\pm$0.000
Caveman ($N=100$)	Glass-Map	0.444 $\pm$ 0.278	0.458 $\pm$ 0.273
	Glass-SE†	0.882$\pm$0.000	0.837$\pm$0.000
	Louvain	1.000$\pm$0.000	1.000$\pm$0.000
	Leiden	1.000$\pm$0.000	1.000$\pm$0.000
	Infomap	1.000$\pm$0.000	1.000$\pm$0.000
	Spectral	1.000$\pm$0.000	1.000$\pm$0.000
	deDoc†	–	–
	HCSE†	0.421 $\pm$ 0.000	0.733 $\pm$ 0.000
	SEClust-Auto†*	1.000$\pm$0.000	1.000$\pm$0.000
	SEClust-CK†*	0.889 $\pm$ 0.000	0.969 $\pm$ 0.000
	SEClust-ML†*	1.000$\pm$0.000	1.000$\pm$0.000
	SEP†	1.000$\pm$0.000	1.000$\pm$0.000
SBM Clean ($N=150$)	LSENet†	0.956 $\pm$ 0.048	0.988 $\pm$ 0.013
	Glass-Mod	0.852 $\pm$ 0.040	0.956 $\pm$ 0.014
	Glass-Map	0.786 $\pm$ 0.000	0.929 $\pm$ 0.000
	Glass-SE†	0.893 $\pm$ 0.057	0.969 $\pm$ 0.018
	Louvain	1.000$\pm$0.000	1.000$\pm$0.000
	Leiden	1.000$\pm$0.000	1.000$\pm$0.000
	Infomap	0.996 $\pm$ 0.007	0.994 $\pm$ 0.010
	Spectral	1.000$\pm$0.000	1.000$\pm$0.000
SBM Clean ($N=150$)	deDoc†	–	–
	HCSE†	0.984 $\pm$ 0.013	0.978 $\pm$ 0.017
	SEClust-Auto†*	0.562 $\pm$ 0.039	0.737 $\pm$ 0.020

Dataset	Algorithm	ARI	NMI
	SEClust-CK ^{†*}	0.992±0.014	0.988±0.021
	SEClust-ML ^{†*}	0.611±0.052	0.767±0.022
	SEP [†]	0.949±0.036	0.932±0.048
	LSENet [†]	1.000±0.000	1.000±0.000
	Glass-Mod	0.996±0.007	0.994±0.010
	Glass-Map	0.682±0.232	0.764±0.172
	Glass-SE [†]	0.996±0.007	0.994±0.010
SBM Noisy ($N=150$)	Louvain	0.080±0.031	0.129±0.022
	Leiden	0.083±0.028	0.131±0.020
	Infomap	0.000±0.000	0.000±0.000
	Spectral	0.095±0.045	0.140±0.038
	deDoc [†]	—	—
	HCSE [†]	0.055±0.036	0.058±0.033
	SEClust-Auto ^{†*}	0.051±0.017	0.119±0.029
	SEClust-CK ^{†*}	0.085±0.028	0.088±0.023
	SEClust-ML ^{†*}	0.050±0.035	0.120±0.046
	SEP [†]	0.064±0.027	0.078±0.026
	LSENet [†]	0.118±0.056	0.116±0.048
	Glass-Mod	0.136±0.096	0.133±0.081
	Glass-Map	0.090±0.062	0.138±0.054
	Glass-SE[†]	0.138±0.106	0.135±0.091

Note on deDoc: The reference agglomerative implementation is $O(N^3)$ per run; for Karate ($N=34$) each seed takes ~ 5.6 s. For $N>50$ the runtime is prohibitive and results are marked —. An $O(m \log N)$ online update (as in the original paper) would resolve this limitation. **Note on HCSE:** The companion library’s `hierarchical_se_clustering` builds a binary coding tree (`target_k=2`); for $K>2$ grounds truth the tree undercounts communities, reducing ARI (e.g., Caveman $K=10$: ARI 0.421). The original Pan et al. [38] construction is depth-unlimited and achieves higher accuracy. **Note on SEClust-Auto vs SEClust-CK:** SEClust-Auto finds K automatically; SEClust-CK uses ground-truth K . Because the SBM signal-to-noise is low, even with the correct K all heuristics score at the detection threshold on SBM-Noisy.

6.3 Results: Objective Evaluators

Table 3 cross-evaluates every algorithm’s output against all three clustering objectives.

6.4 Results: Real-World Attributed Graphs

Table 4 reports results on the two standard attributed citation networks used in the LSENet paper [53]. Crucially, the ground-truth partition in these datasets is *semantic* (paper topics) rather than topological (citation density).

Topology-only vs. feature-augmented. On feature-rich datasets, feature-augmented LSENet outperforms topology-only Glass-SE in ACC (Cora: $0.387 > 0.274$; Citeseer: $0.403 > 0.252$), because the ground-truth classes (academic subjects) are poorly aligned with citation-graph topology. Louvain dominates on NMI, reflecting that the citation structure still carries significant mutual information with ground truth, even if it does not fully separate classes. This is consistent with prior observations [53, 8] that topology-only SE objectives are best suited to graphs where cluster boundaries coincide with connectivity bottlenecks. Incorporating node features via a GNN encoder — as in LSENet — is the natural path to extending structural entropy minimization to attributed graphs.

6.5 Analysis and Takeaways

Noisy graph superiority. On the challenging **SBM (Noisy)** dataset (signal near the community-detection threshold), Glass-SE achieved ARI 0.138, the highest among all tested methods. All other methods — Louvain (0.080), Leiden (0.083), Infomap (0.000), Spectral (0.095), HCSE (0.055), SEClust-CK (0.085), SEClust-ML (0.050), SEP (0.064), LSENet (0.118), Glass-Mod (0.136) — fall at or below Glass-SE on this hardest regime. This demonstrates the resilience of direct soft-logit optimization via `vmap` under high noise; the continuous relaxation effectively averages over multiple near-optimal partitions.

Clean graphs: K -constrained SE dominates. On **SBM (Clean)** and **Karate**, SEClust-CK (ARI 0.992 and 0.882) and HCSE (0.984 and 0.882) match or nearly match the best achievable performance, confirming that SE minimization with correct K is a reliable and accurate clustering criterion when the signal is strong.

Caveman: free- K multi-level wins. On the Caveman graph (10 perfectly-disjoint cliques), SEClust-ML and all discrete baselines achieve ARI 1.000, while HCSE drops to 0.421. The failure of the binary coding tree (HCSE) on $K=10$ reflects the mismatch between binary tree structure and a K -way ground truth. Free- K multi-level search naturally discovers all 10 components.

Caveman limitation for continuous methods. The only setting where discrete methods (ARI 1.000) clearly exceeded Glass-SE (0.893) was the Caveman graph, whose perfectly disjoint cliques present a degenerate topology. Because there are literally zero inter-clique edges, no gradient signal propagates to push cluster assignments apart; continuous relaxation struggles to represent absolute zero probabilities. An L1 sparsity penalty or feature projection layer (similar to LSENet’s MLP) resolves this, but is unnecessary for real-world, dense, noisy graphs.

K -constraint artifact. On Karate, Louvain achieves a lower (better) raw SE objective (3.323) than Glass-SE (3.731) because Louvain freely optimizes K (choosing $K=4$) while accuracy-benchmarked Glass-SE is constrained to $K=2$. When Glass-SE is allowed to optimize freely over $K=8$, it finds SE 3.245, beating Louvain’s global minimum.

Companion library. All experimental code and configurations are released in `glass-jax` at <https://github.com/SELGroup/glass-jax>. See Appendix A for reproduction instructions.

Runtime and scalability. The benchmark also highlights a practical scalability divide. Discrete baselines such as Louvain and Leiden remain the fastest choices for large sparse graphs because they operate directly on sparse neighborhoods. Reference deDoc-style agglomeration is cubic in its naive form and is therefore reported only for Karate; in our implementation each Karate seed takes about 5.6 seconds, and $N>50$ runs are marked infeasible. The differentiable Glass objectives are optimized by dense soft assignments over 500 Adam steps and four vectorized starts, making them most appropriate for medium-scale graphs or neural latent graphs where gradient flow matters. Scaling differentiable SE to million-node graphs will require sparse kernels, mini-batch approximations, or coarsen-and-refine objectives, which we identify as a major open problem in Section 8.

Benchmark scope and open problem O7. This benchmark covers representative topology-only methods on synthetic graphs and real-world attributed graphs, but several important method classes remain outside its scope. Hierarchical SE methods (HCSE hierarchy depth, Paris, Agglomerative) require a dedicated dendrogram quality metric (e.g., Dasgupta cost), which we defer to future work (see Open Problem O7, §8). Scalability evaluation on million-node graphs (LFR benchmarks, ogbn-arxiv) is likewise deferred, pending `glass-jax` sparse-matrix support. The authors encourage the community to contribute standardized SE benchmark results to <https://github.com/SELGroup/glass-jax/benchmarks>.

7 Discussion

The decade of work surveyed above suggests that SE is best understood not as a single clustering score, but as a unifying principle for graph intelligence: systems become intelligible when their uncertainty can be reduced by a compact hierarchy. This interpretation explains why SE appears in apparently distant settings, including graph pooling, chromatin-domain discovery, event detection, social bot analysis, and reinforcement learning. In each case, the method searches for a structural abstraction that preserves the dominant organization while suppressing incidental variation.

For TGINA’s focus on graph intelligence and network applications, SE is especially useful because it connects three concerns that are often handled separately. First, it is a *measure*: the entropy value summarizes how ordered or disordered a network is under a candidate hierarchy. Second, it is an *optimizer*: minimizing the measure constructs the hierarchy itself. Third, it is a *learning signal*: differentiable variants allow the same principle to shape GNN encoders, pooling layers, and decision abstractions. This measure–optimizer–learning-signal triad is the core reason SE deserves a full journal treatment rather than a short method catalogue.

At the same time, the benchmark results caution against overclaiming. Topology-only SE objectives are most appropriate when communities correspond to structural bottlenecks. On attributed citation networks, semantic labels may be better explained by text features than by citation topology, and feature-augmented models such as LSENet can therefore dominate in accuracy. Conversely, on noisy synthetic topologies, differentiable SE remains competitive with or stronger than standard baselines. The practical lesson is that SE should be matched to the source of structure: pure topology for latent abstraction and network organization, feature-aware SE for semantic graph learning, and constrained SE for domain-specific structures such as genomic intervals.

Finally, SE offers a promising language for trustworthy graph intelligence in LLM and autonomous-agent systems. GraphRAG pipelines, tool-use traces, memory graphs, and multi-agent interaction networks all produce evolving relational structures whose reliability depends on whether useful organization can be separated from noisy connectivity. SE can quantify that organization, detect disorder, and potentially guide the construction of stable retrieval or decision hierarchies. This connection remains early, but it gives a concrete path from classical structural information theory to the emerging graph intelligence problems that motivate the TGINA venue.

8 Open Problems and Future Directions

Despite a decade of rapid progress, several fundamental challenges remain open:

O1. Unified Framework for Complex Network Types

Existing SE variants have been developed independently for directed graphs, hypergraphs, multi-relational graphs, and dynamic networks. A pressing open problem is the development of a *unified SE framework* that handles these structures under a single formalism. This would require extending the encoding-tree formulation to support non-standard volume and cut measures, and establishing formal connections between the specialized variants.

O2. Scalable SE Minimization

The dominant heuristic SE algorithms scale as $O(n^2)$, limiting applicability to graphs with hundreds of thousands of nodes. Although SSSE achieves nearly linear complexity under a sampling approximation, theoretical guarantees on the approximation quality remain weak. Developing provably good approximation algorithms for SE minimization, analogous to those for modularity and the map equation, is an important open direction.

O3. Differentiable SE for Deep Learning Integration

Current differentiable SE formulations (LSENet, DeSE, Glass-SE) are restricted to 2-level (flat) trees. Extending differentiable SE to arbitrary-depth hierarchical trees—while maintaining well-defined gradients and computational tractability—is an open problem. This would enable end-to-end learning of hierarchical representations in GNNs and Transformers.

O4. SE and Large Language Models

The recent success of LLMs raises the question of whether SE can be used to analyse or improve the latent graph structures induced by LLM attention patterns and knowledge graphs. Preliminary connections have been drawn via SE-based knowledge quantification [63] and graph-augmented reasoning [26], but a principled SE-based theory of LLM structure is absent.

O5. Theoretical Connections to Other Complexity Measures

While [31] established a bound between SE and VNE, the exact relationship between SE and other information-theoretic graph measures (e.g., Kolmogorov complexity, Fisher information, persistent homology) is not well understood. A deeper theoretical investigation would both strengthen SE’s foundations and enable new algorithmic insights.

O6. SE in Partially Observable Environments

SE-based RL methods (SISA, SIRD, SI2E) have been developed for fully observable settings. Extending SE to partial observability—where the agent’s state graph is uncertain or must be inferred—is an important frontier for real-world applicability.

O7. Benchmark Standardization

A major bottleneck for the field is the absence of a standardized evaluation protocol. Different papers use different datasets, train–test splits, and evaluation metrics, making direct comparison difficult. We call for the community to adopt a standard benchmark suite (as partially initiated in Section 6) with open, reproducible implementations.

O8. SE for Fairness and Privacy

The use of SE to deliberately obscure community structure [32] suggests promising applications in *privacy-preserving graph publishing* and *fairness-aware graph learning*. A principled theory connecting SE-based obfuscation to differential privacy guarantees remains an open problem.

9 Conclusion

This survey has systematically charted the landscape of Structural Entropy research over the decade since its introduction in 2016. We have consolidated its theoretical underpinnings, computational methodologies, and burgeoning applications across AI, presenting the field’s first comprehensive synthesis.

Our review reveals that SE offers a distinct and powerful lens for artificial intelligence: it shifts the focus from local, pairwise interactions to the global, hierarchical organization of complex systems. We have detailed how minimizing SE serves as a constructive principle to uncover a system’s most efficient information-theoretic abstraction—a capability that has been effectively harnessed in graph learning for pooling and structure augmentation, in reinforcement learning for hierarchical state–action abstraction and exploration, and in diverse application domains from genomics to speech processing.

The reproducibility benchmark in Section 6 demonstrates that differentiable SE objectives, as implemented in the **glass-jax** library, match or exceed state-of-the-art methods on realistic, noisy graph topologies while offering the key advantage of end-to-end gradient-based optimization.

In conclusion, Structural Entropy provides more than a new set of tools: it offers a principled way of thinking about information and structure in a hierarchical world. By providing a rigorous method to quantify and optimize organizational uncertainty within data, SE holds the potential to guide the development of AI systems that are not only more powerful but also more interpretable, robust, and theoretically grounded. The eight open problems identified in Section 8 chart a research agenda for the next decade.

Differences from the IJCAI-2025 Conference Version. This journal article substantially extends [51] in the following ways:

1. A new, self-contained *formal foundations* section with precise definitions and comparative analysis against modularity, map equation, and von Neumann entropy.
2. An extended *algorithm catalogue* (Table 1) covering all known SE minimization methods with unified complexity notation.

3. A new *reproducibility benchmark* (Section 6) with empirical comparisons over multiple datasets and random seeds.
4. Expanded *application coverage*: speech processing (SECodec, SEVC), geoscience knowledge quantification (KQI), and updated 2025–2026 literature.
5. A structured *open problems* section (Section 8) absent from the conference version.
6. Release of `glass-jax` as a companion reproducibility library.

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A Survey Protocol and Companion Library

A.1 Literature Search Protocol

This survey covers SE-related publications from January 2016 (the publication date of the foundational SE paper [16]) through May 2026. We conducted systematic searches on Google Scholar, Semantic Scholar, arXiv, and the ACL/ACM/IEEE digital libraries using the following query terms: “structural entropy”, “encoding tree”, “SE minimization”, “hierarchical structural entropy”, and “Li Angsheng”. We included papers that (a) directly use SE as a primary methodology, (b) propose extensions of the SE formalism, or (c) introduce SE-based applications. Papers using the term “structural entropy” in a purely figurative sense without connection to the Li–Pan framework [16] were excluded.

A.2 The `glass-jax` Companion Library

`glass-jax` (Graph-based Latent Abstraction & Structural Segmentation in JAX) is a fully differentiable, hardware-accelerated framework for graph clustering that serves as this survey’s reproducibility artifact.

Core objectives.

- **Soft Modularity:** relaxes the Louvain objective via $Q = \frac{1}{2m} \text{Tr}(S^\top BS)$ with soft assignment matrix S .
- **Soft Map Equation:** a differentiable approximation of Infomap computing the description length of a random walk over soft assignments.
- **Soft 2D Structural Entropy (Glass-SE):** soft-assignment implementation of Definition 3.2, enabling gradient flow through the SE objective.

Optimization engine. The *multi-start vmap engine* runs B parallel optimization trajectories using `jax.vmap`, initialized with 1 spectral embedding and $B-1$ random Gaussian initializations. Temperature annealing decays τ from 1.0 to 0.01 over 500 Adam steps,

sharpening soft assignments into near-discrete decisions. The trajectory achieving the best (unscaled) objective value is selected.

Benchmarking interface. All experiments in Section 6 are fully reproducible via:

```
git clone https://github.com/SELGroup/glass-jax
cd glass-jax
pip install -e ".[dev]"
python experiments/benchmark_same_protocol.py
```

Requirements. $\text{JAX} \geq 0.4$, `optax`, `networkx`, `scikit-learn`, `python-louvain`, `infomap` (C++ wrapper). GPU/TPU acceleration is automatic via JAX's XLA compiler.

Table 3: Cross-objective evaluation (mean $\pm$ 95% CI over 5 seeds). Modularity $\uparrow$; Map Equation and Structural Entropy $\downarrow$. Bold: best per column per dataset.

Dataset	Algorithm	Modularity $\uparrow$	Map Eq. $\downarrow$	Struct. Ent. $\downarrow$
Karate	Louvain	0.440$\pm$0.006	4.165 $\pm$ 0.044	3.323$\pm$0.015
	Infomap	0.435 $\pm$ 0.000	4.087$\pm$0.000	3.495 $\pm$ 0.000
	SEP	0.404 $\pm$ 0.000	4.198 $\pm$ 0.000	3.731 $\pm$ 0.000
	LSENet	0.404 $\pm$ 0.000	4.198 $\pm$ 0.000	3.731 $\pm$ 0.000
	Glass-Mod	0.404 $\pm$ 0.000	4.198 $\pm$ 0.000	3.731 $\pm$ 0.000
	Glass-Map	0.224 $\pm$ 0.083	4.680 $\pm$ 0.193	3.983 $\pm$ 0.135
	Glass-SE	0.404 $\pm$ 0.000	4.198 $\pm$ 0.000	3.731 $\pm$ 0.000
Caveman	Louvain	0.878$\pm$0.000	3.548$\pm$0.000	3.394$\pm$0.000
	Infomap	0.878$\pm$0.000	3.548$\pm$0.000	3.394$\pm$0.000
	SEP	0.878$\pm$0.000	3.548$\pm$0.000	3.394$\pm$0.000
	LSENet	0.870 $\pm$ 0.008	3.623 $\pm$ 0.080	3.471 $\pm$ 0.083
	Glass-Mod	0.852 $\pm$ 0.008	3.807 $\pm$ 0.075	3.664 $\pm$ 0.081
	Glass-Map	0.824 $\pm$ 0.000	3.972 $\pm$ 0.000	3.827 $\pm$ 0.000
	Glass-SE	0.860 $\pm$ 0.010	3.729 $\pm$ 0.098	3.585 $\pm$ 0.106
SBM (Clean)	Louvain	0.418$\pm$0.006	6.909$\pm$0.022	6.011$\pm$0.009
	Infomap	0.418$\pm$0.006	6.909$\pm$0.022	6.011$\pm$0.009
	SEP	0.408 $\pm$ 0.010	6.945 $\pm$ 0.037	6.026 $\pm$ 0.015
	LSENet	0.418$\pm$0.006	6.909$\pm$0.022	6.011$\pm$0.009
	Glass-Mod	0.418$\pm$0.006	6.909$\pm$0.022	6.011$\pm$0.009
	Glass-Map	0.317 $\pm$ 0.072	7.112 $\pm$ 0.149	6.275 $\pm$ 0.191
	Glass-SE	0.418$\pm$0.006	6.909$\pm$0.022	6.011$\pm$0.009
SBM (Noisy)	Louvain	0.212$\pm$0.007	7.758 $\pm$ 0.045	6.202$\pm$0.005
	Infomap	0.000 $\pm$ 0.000	7.187$\pm$0.004	7.187 $\pm$ 0.004
	SEP	0.167 $\pm$ 0.007	7.757 $\pm$ 0.023	6.400 $\pm$ 0.015
	LSENet	0.204 $\pm$ 0.009	7.652 $\pm$ 0.031	6.337 $\pm$ 0.016
	Glass-Mod	0.205 $\pm$ 0.011	7.644 $\pm$ 0.036	6.337 $\pm$ 0.020
	Glass-Map	0.128 $\pm$ 0.038	7.562 $\pm$ 0.079	6.623 $\pm$ 0.133
	Glass-SE	0.208 $\pm$ 0.011	7.638 $\pm$ 0.036	6.330$\pm$0.018

Table 4: Clustering on real-world citation networks (ACC, NMI, ARI). †: topology-only method; ‡: feature-augmented method. Best per metric per dataset in bold.

Dataset	Algorithm	ACC	NMI	ARI
Cora ($N=2,708$, $K=7$)	Louvain [†]	0.372	0.439	0.236
	Glass-SE [†]	0.274	0.076	0.039
	LSENet [‡]	0.387	0.266	0.164
Citeseer ($N=3,327$, $K=6$)	Louvain [†]	0.192	0.329	0.094
	Glass-SE [†]	0.252	0.037	0.025
	LSENet [‡]	0.403	0.195	0.166